

Victor Sechaud

PHD STUDENT · APPLY MATHEMATICS



About me

I am a PhD student in Applied Mathematics at ENS de Lyon, supervised by Julián Tachella and Patrice Abry. I'm working on inverse problems and machine learning, in particular, self-supervised learning methods for non-linear problems such as phase retrieval and declipping.

Interests

- Machine learning
- Deep learning
- Optimization
- Statistics
- Compressive sensing
- Zero-order optimization

Contact

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Personal

Nationality: French
Date of birth: 15.11.1999

Languages

French	Native
English	B2+

🌐 [vsechaud.github.io/](https://github.com/vsechaud)

🔗 <https://github.com/vsechaud>

EDUCATION

- 2023-now **PhD in Apply Mathematics**
CNRS, ENS DE LYON,
Thesis: "Self-supervised learning for non-linear inverse problems" 📍 LYON, FRANCE
- 2023 **Master's degree**
UNIVERSITY LYON 1 CLAUDE BERNARD
Apply mathematics 📍 LYON, FRANCE
- 2022 **Master's degree**
UNIVERSITY LYON 1 CLAUDE BERNARD
Agrégation de mathématiques (Highly selective French national teaching qualification) 📍 LYON, FRANCE
- 202 **Undergraduate degree**
UNIVERSITY LYON 1 CLAUDE BERNARD
Mathematics 📍 LYON, FRANCE

ACADEMIC EXPERIENCE

- 2025 sept-now **Visiting scholar**
HERIOT-WATT UNIVERSITY
9-week visiting researcher with Marcelo Pereyra 📍 EDINBURGH, UK
- 2023-now **Teaching**
ENS DE LYON
Mathematics department: Linear Algebra, Python, and Statistics 📍 LYON, FRANCE

PUBLICATIONS

- 2026 J. Tachella, **V. Sechaud**, L. Jacques, P. Abry, and J. Tachella, "Learning to reconstruct from saturated data: audio declipping and high-dynamic range imaging", *TMLR*, 2026.
V. Sechaud, J. Scanvic, Q. Barthélemy, P. Abry, and J. Tachella, "Equivariant Splitting: Self-supervised learning from incomplete data", *ICLR*, 2026.
- 2025 J. Tachella, L. Davy, J. Scanvic, **V. Sechaud**, R. Vo, *et al*, "DeepInverse: A Python package for solving imaging inverse problems with deep learning", *Journal of Open Source Software*, 2025.
V. Sechaud, P. Abry, L. Jacques, and J. Tachella, "Apprentissage auto-supervisé pour la reconstruction de phase", *GRETSI*, 2025.
- 2024 **V. Sechaud**, L. Jacques, P. Abry, and J. Tachella, "Equivariance-based self-supervised learning for audio signal recovery from clipped measurements", *EUSIPCO 2024*, 2024.

TALKS

- 2026 **Equivariant splitting: self-supervised learning method**
ICLR (POSTER) 📍 RIO DE JANEIRO, BRAZIL
- 2026 **Reconstruct Anything Model (RAM)**
ICLR (POSTER) 📍 RIO DE JANEIRO, BRAZIL
- 2025 **Combining equivariance with splitting methods**
HERIOT-WATT UNIVERSITY 📍 EDINBURGH, UK
- 2025 **Self-supervised learning for phase retrieval problem**
GRETSI 📍 STRASBOURG, FRANCE
- 2024 **learning to reconstruct from clipped signals alone**
EUSIPCO 📍 LYON, FRANCE